

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off020GH001minFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off020GH001minFF
```

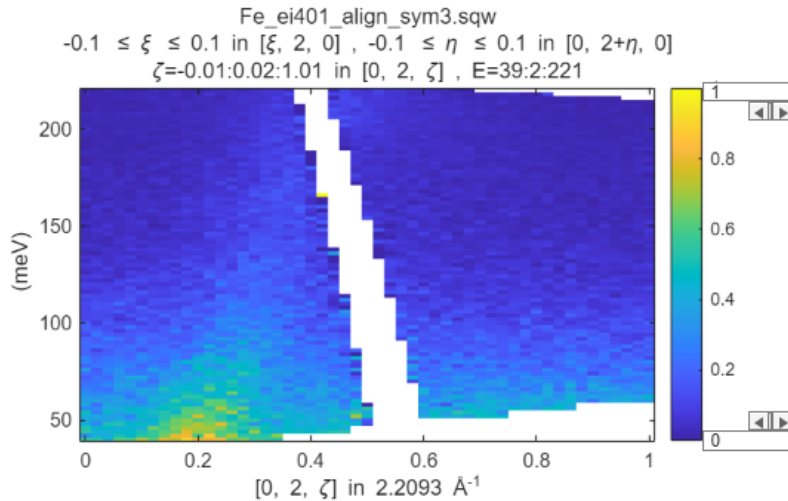
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

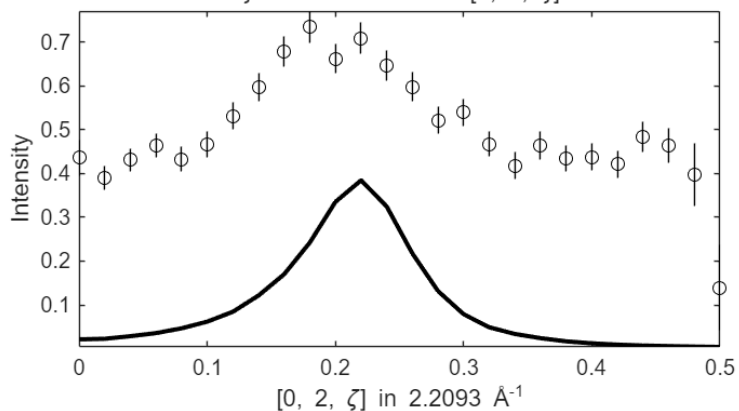
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFcutsFits.(selectCuts) = cuts_list;

GH_FFcutsFits = fit_multicutsFF_along_direction(...
    GH_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

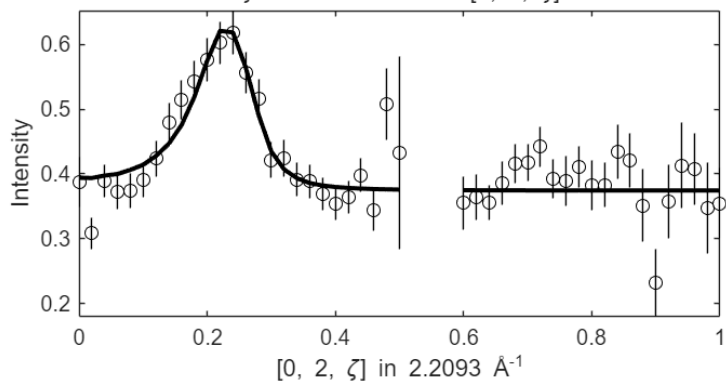
```
*****
**** En = 45±5 going to 215
*****
```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $40 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

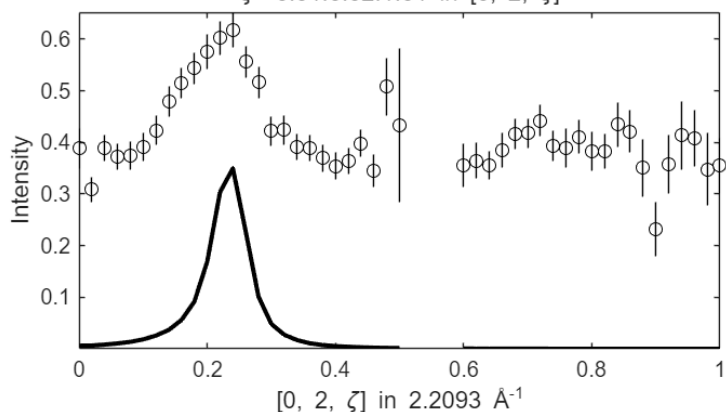


**** En = 55±5 going to 215

Fe_ei401_align_sym3.sqw
 $S=0.65\pm0.11$; $J_0= 25\pm0.86$; $J_1= 0\pm 0$; $\gamma=10.8111\pm3.90406$
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $50 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



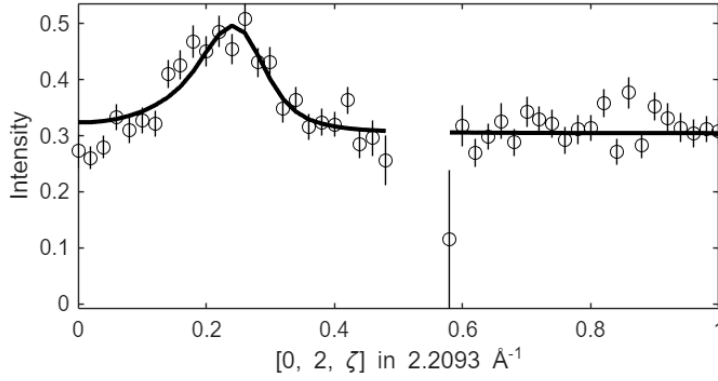
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $50 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



**** En = 65±5 going to 215

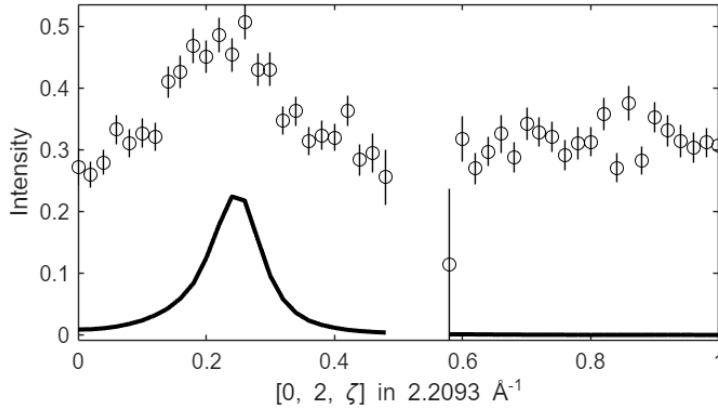
Fe_ei401_align_sym3.sqw

S=0.77±0.16; J0= 27± 1.6; J1= 0± 0; gamma=22.2573±7.86555
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $60 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

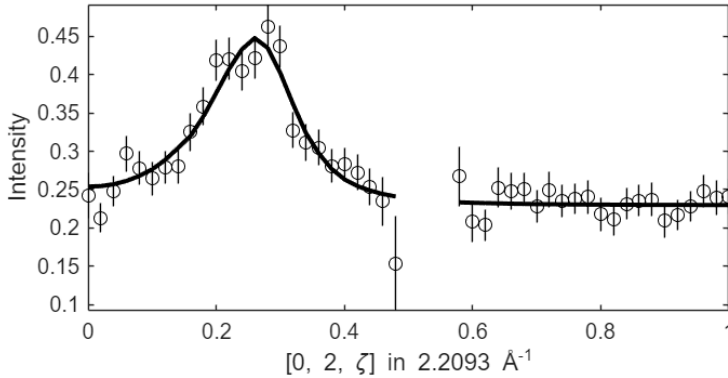
$-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $60 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



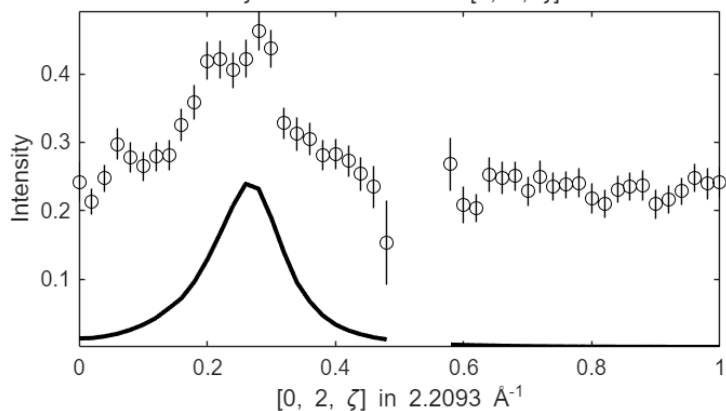
**** En = 75±5 going to 215

Fe_ei401_align_sym3.sqw

S= 1.3±0.05; J0= 29±0.67; J1= 0± 0; gamma=38.5135±1.18975
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $70 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

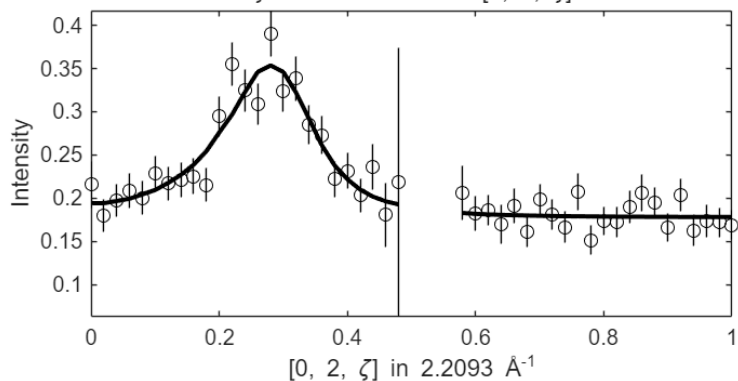


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $70 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

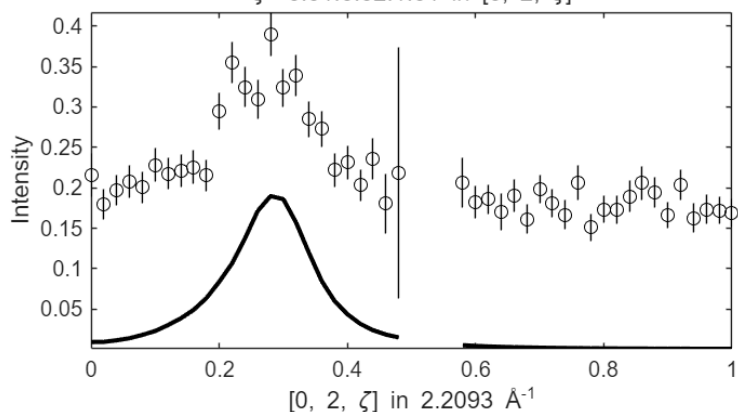


**** En = 85±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.054$; $J_0 = 29 \pm 0.76$; $J_1 = 0 \pm 0$; $\gamma = 43.9605 \pm 1.68238$
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $80 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



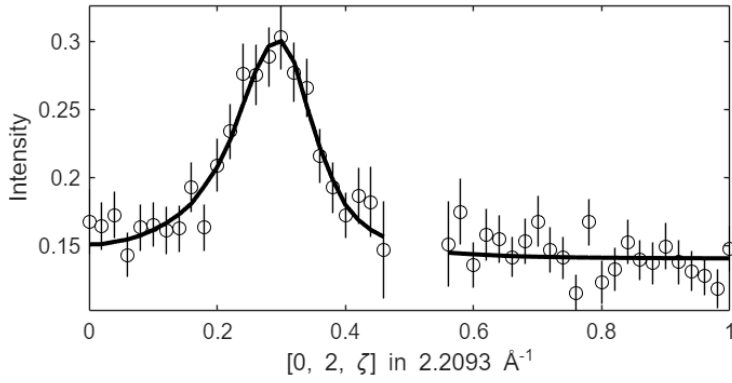
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $80 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



**** En = 95±5 going to 215

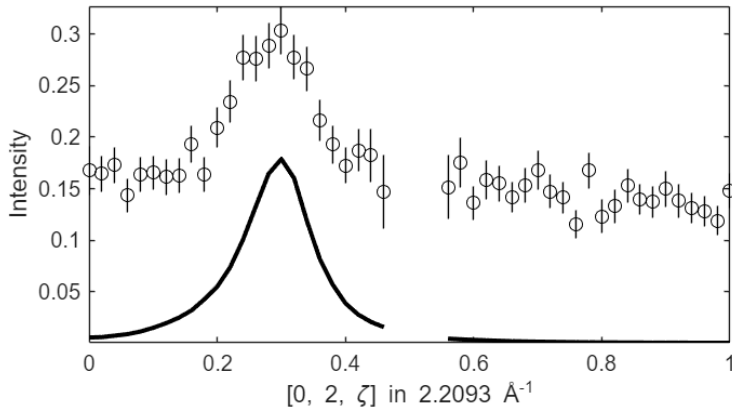
Fe_ei401_align_sym3.sqw

S=0.99±0.11; J0= 30±0.95; J1= 0± 0; gamma=37.8302±6.5216
 $-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $90 \leq E \leq 100$ eV
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

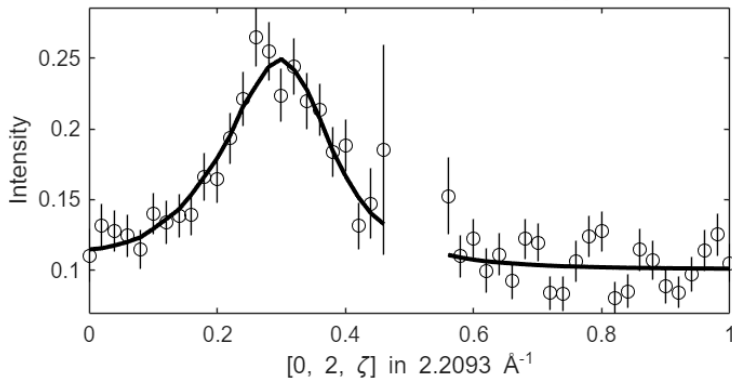
$-0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $90 \leq E \leq 100$ eV
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



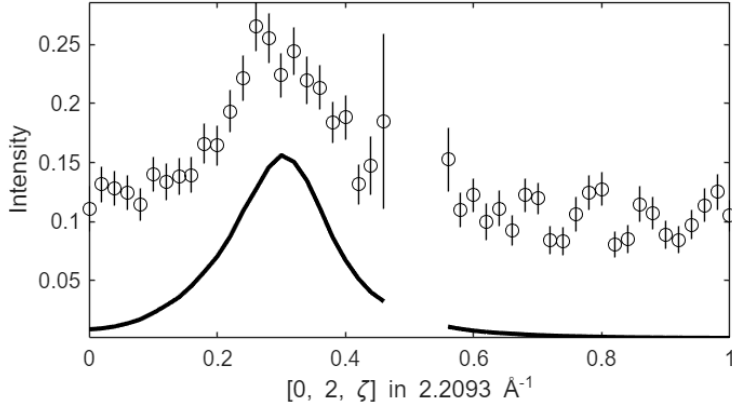
**** En = 105±5 going to 215

Fe_ei401_align_sym3.sqw

S= 1.7±0.33; J0= 35± 2.8; J1= 0± 0; gamma=78.0764±22.6949
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $100 \leq E \leq 110$ eV
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

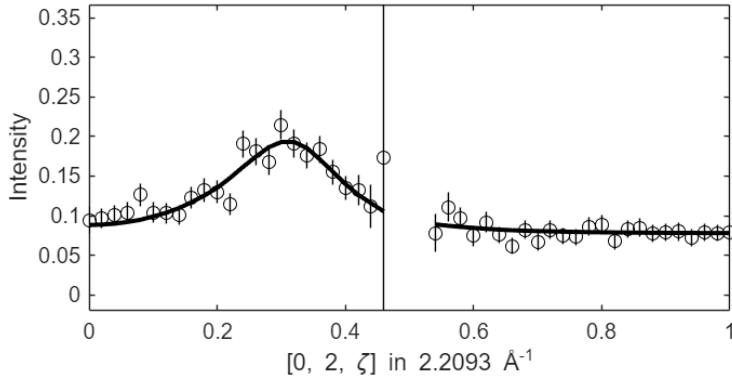


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $100 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

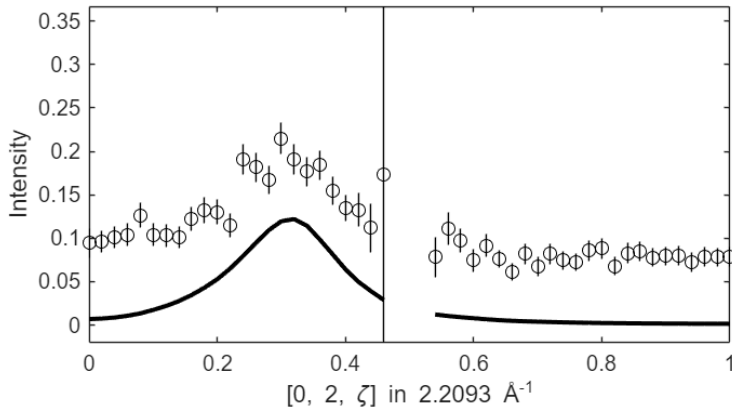


**** En = 115±5 going to 215

Fe_ei401_align_sym3.sqw
 $S= 1.5 \pm 0.31$; $J_0= 37 \pm 2.9$; $J_1= 0 \pm 0$; $\gamma=89.2651 \pm 26.2322$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $110 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



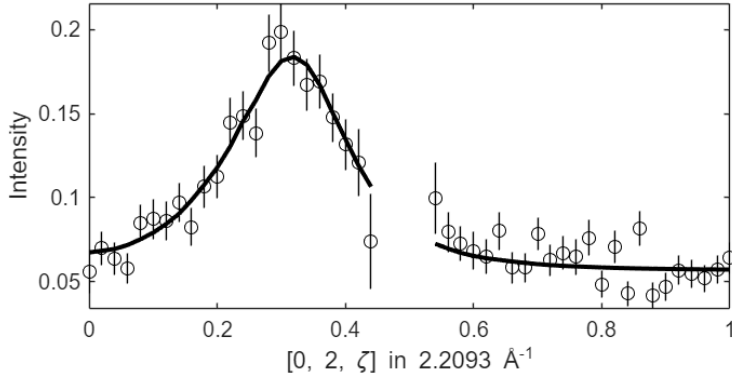
Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $110 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



**** En = 125±5 going to 215

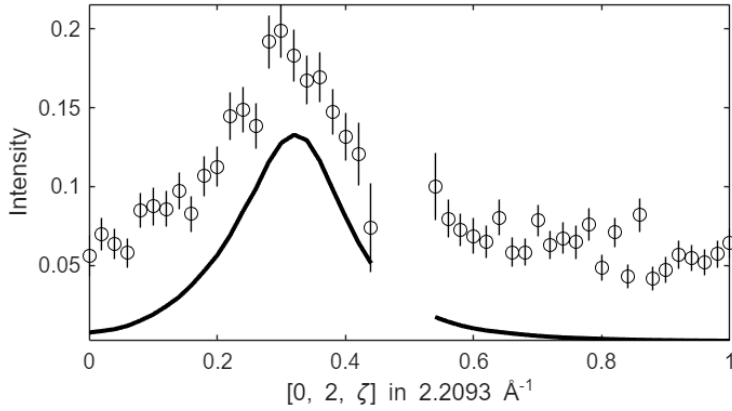
Fe_ei401_align_sym3.sqw

$S = 1.8 \pm 0.4$; $J_0 = 39 \pm 3.3$; $J_1 = 0 \pm 0$; $\gamma = 102.89 \pm 31.8518$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $120 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

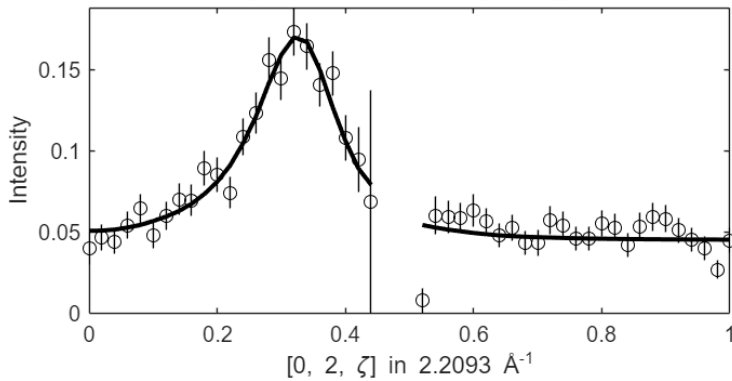
$0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $120 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[0, 2, \zeta]$



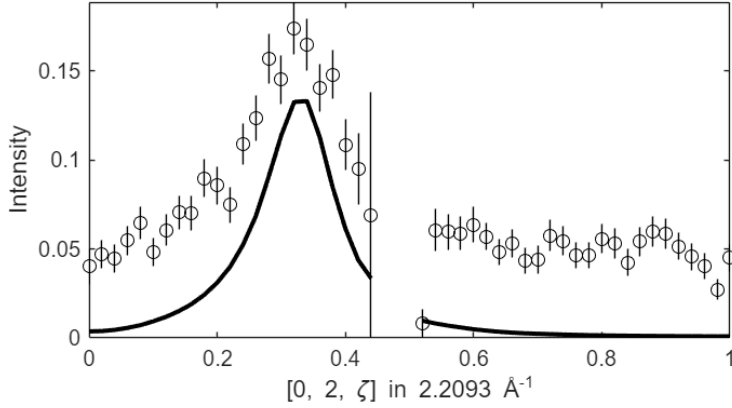
**** En = 135±5 going to 215

Fe_ei401_align_sym3.sqw

$S = 1.1 \pm 0.15$; $J_0 = 35 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 52.9606 \pm 11.5679$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $130 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[0, 2, \zeta]$

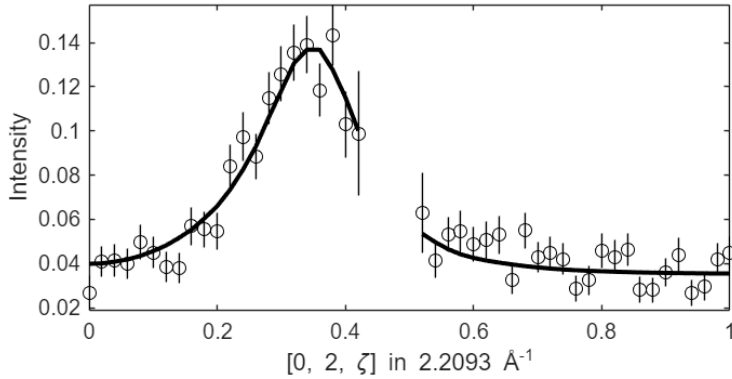


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $130 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

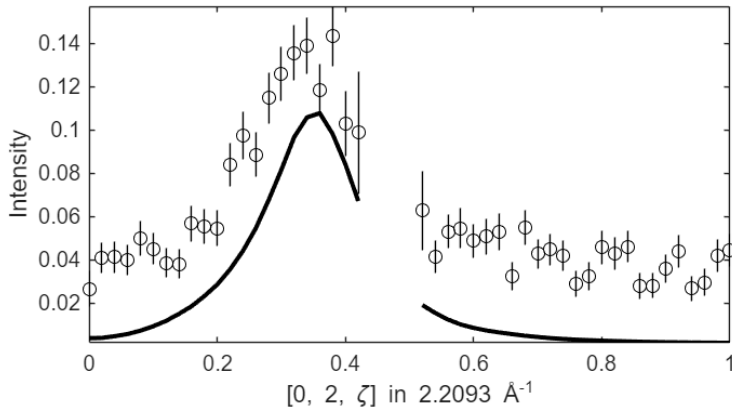


**** En = 145±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.067$; $J_0 = 35 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 77.3069 \pm 2.72405$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $140 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



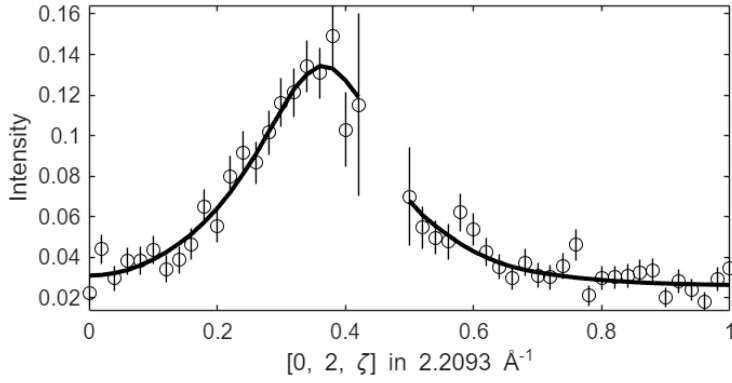
Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $140 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



**** En = 155±5 going to 215

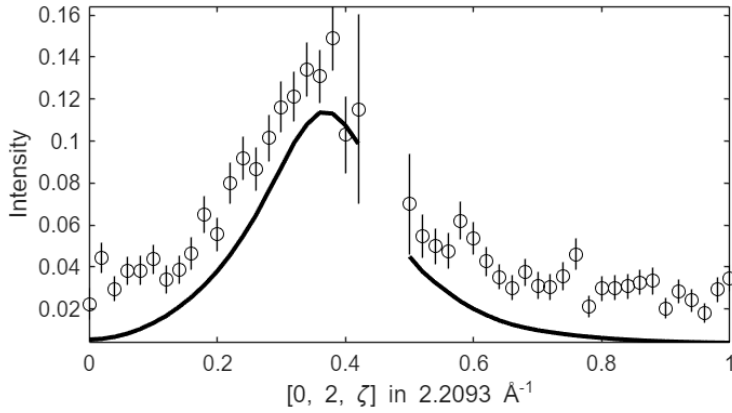
Fe_ei401_align_sym3.sqw

S= 2.1±0.12; J0= 39±0.88; J1= 0± 0; gamma=133.885±4.57402
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $150 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

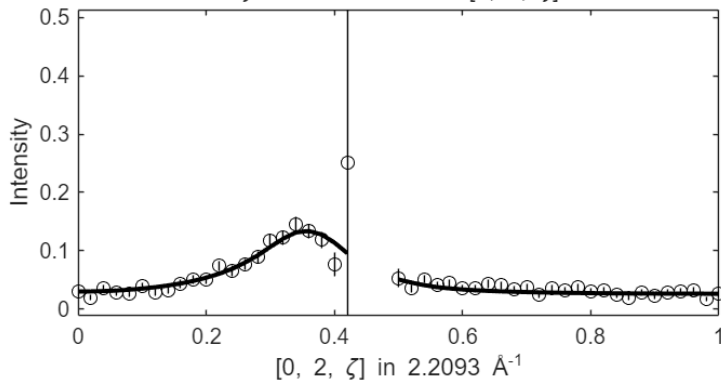
$0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $150 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



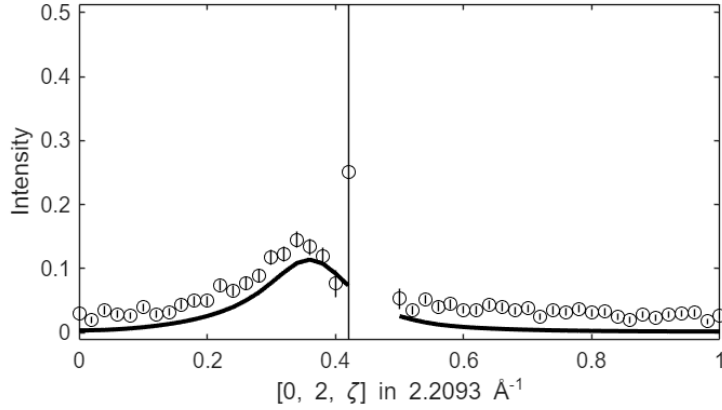
**** En = 165±5 going to 215

Fe_ei401_align_sym3.sqw

S= 1.3±0.16; J0= 38± 1.4; J1= 0± 0; gamma=77.7325±12.6206
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $160 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

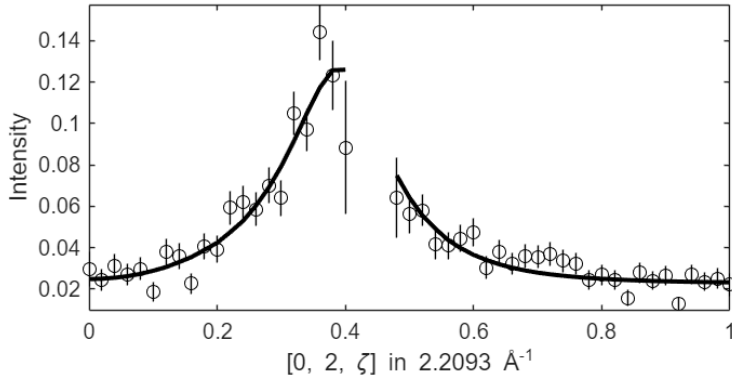


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $160 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

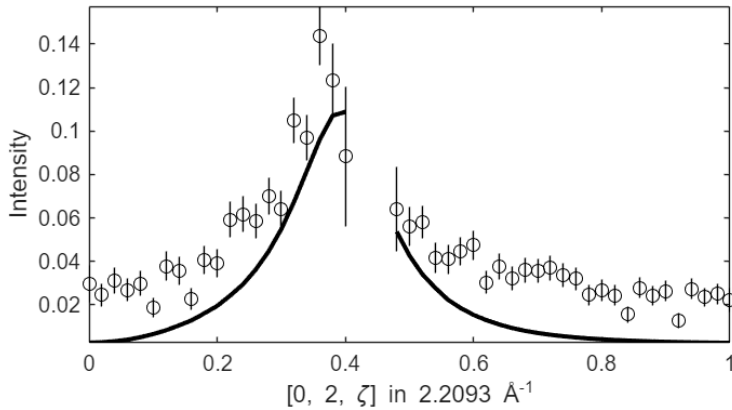


**** En = 175±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.3 \pm 0.087$; $J_0 = 34 \pm 0.79$; $J_1 = 0 \pm 0$; $\gamma = 79.6981 \pm 3.63682$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $170 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $170 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

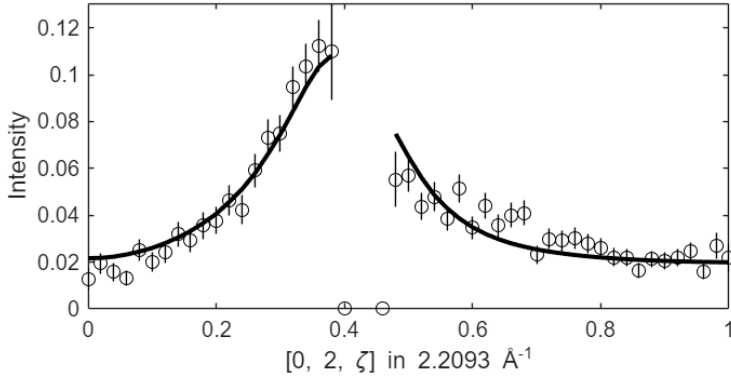


**** En = 185±5 going to 215

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.9216

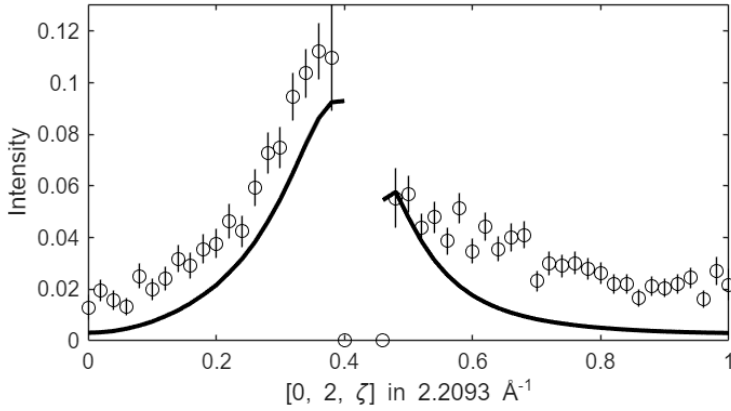
Fe_ei401_align_sym3.sqw

S= 1.5±0.21; J0= 38± 2.6; J1= 0± 0; gamma=108.597±29.0954
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $180 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

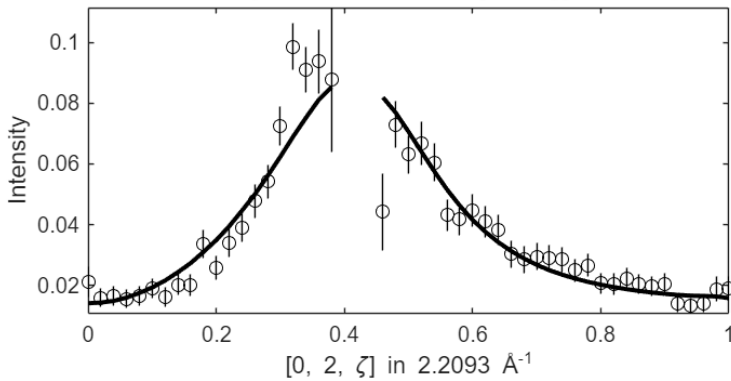
$0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $180 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



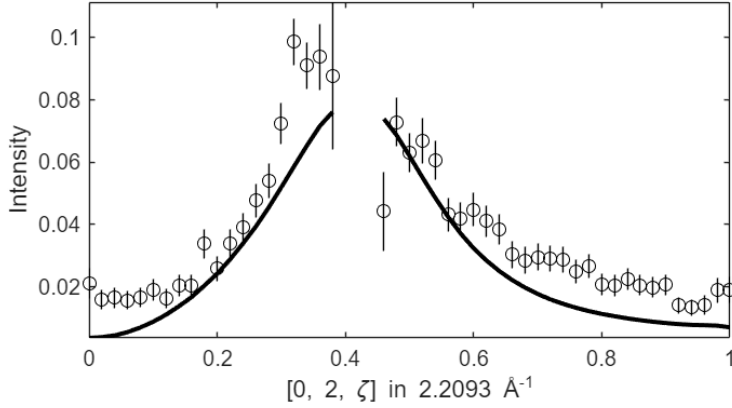
**** En = 195±5 going to 215

Fe_ei401_align_sym3.sqw

S= 2.3±0.59; J0= 42± 5.8; J1= 0± 0; gamma=222.625±86.6958
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $190 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

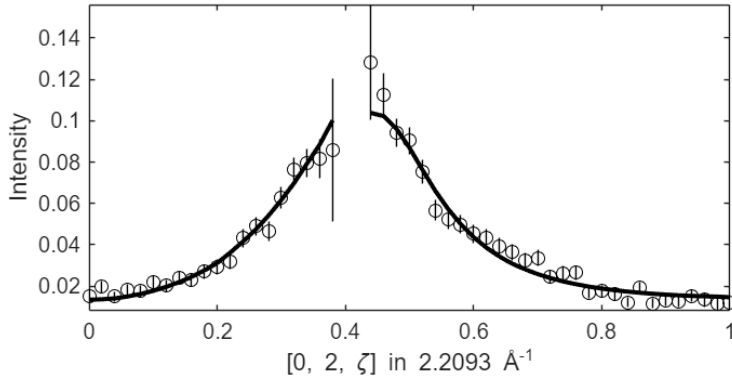


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $190 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

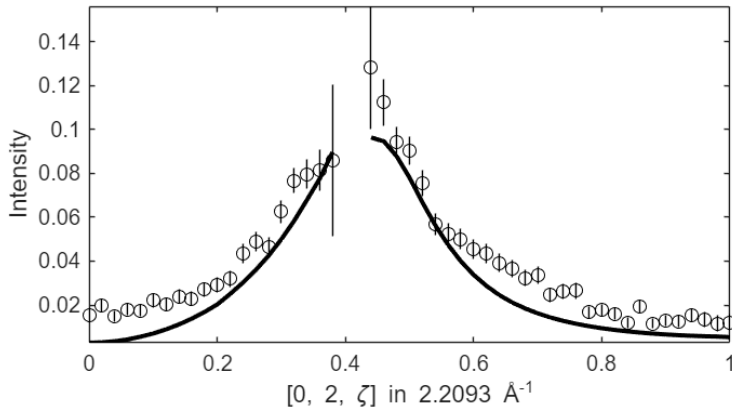


 **** En = 205±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.8 \pm 0.097$; $J_0 = 37 \pm 0.5$; $J_1 = 0 \pm 0$; $\gamma = 130.596 \pm 7.74798$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $200 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $200 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[0, 2, \zeta]$

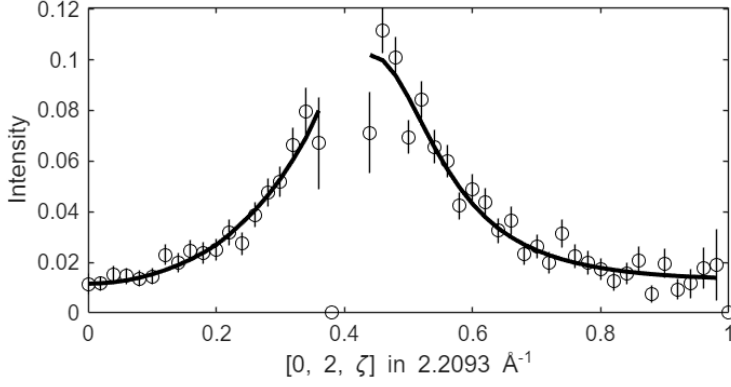


 **** En = 215±5 going to 215

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.9216

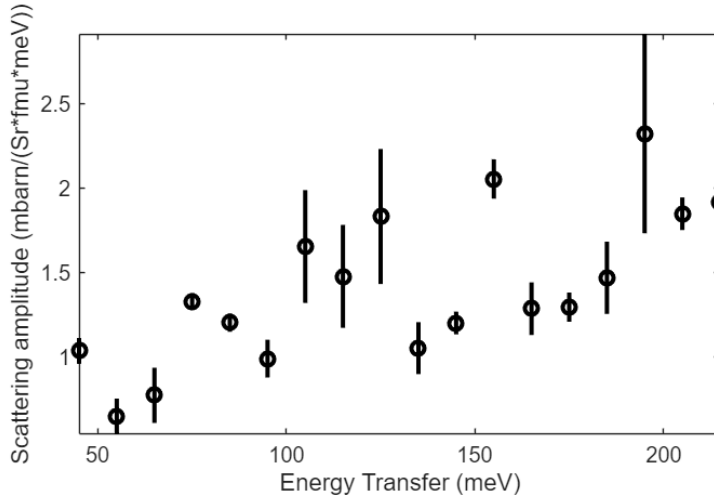
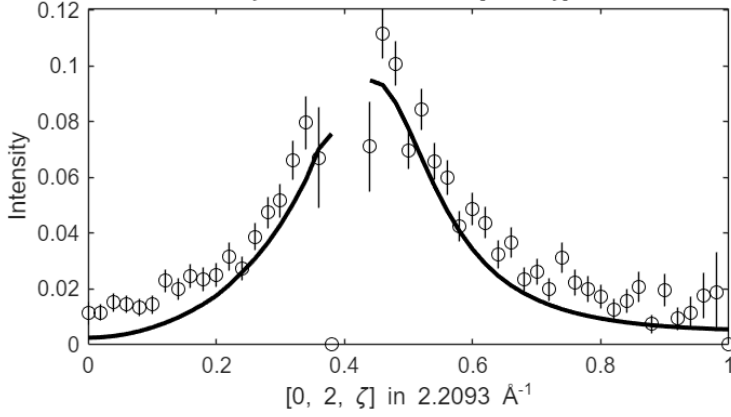
Fe_ei401_align_sym3.sqw

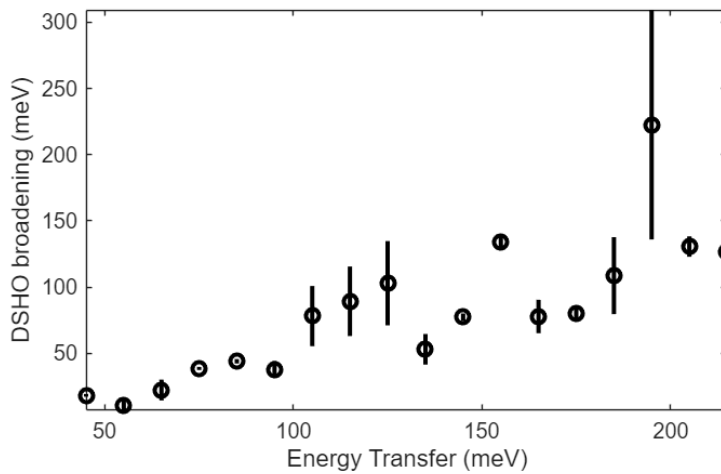
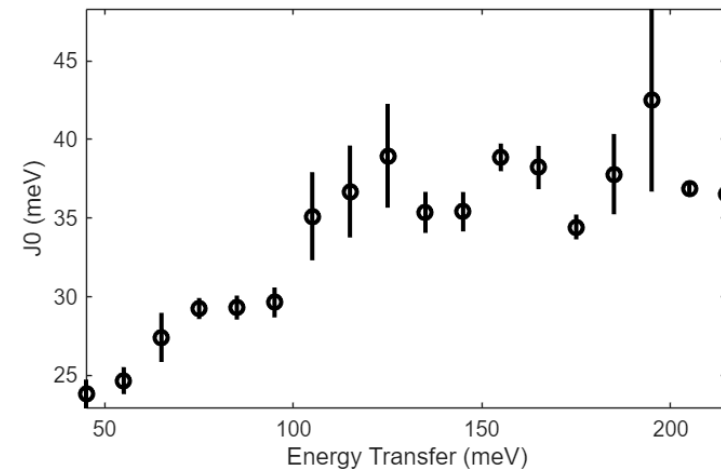
$S = 1.9 \pm 0.17$; $J_0 = 37 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 126.509 \pm 16.3825$
 $0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $210 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[0, 2, \zeta]$



Fe_ei401_align_sym3.sqw

$0.1 \leq \xi \leq 0.1$ in $[\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2+\eta, 0]$, $210 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[0, 2, \zeta]$





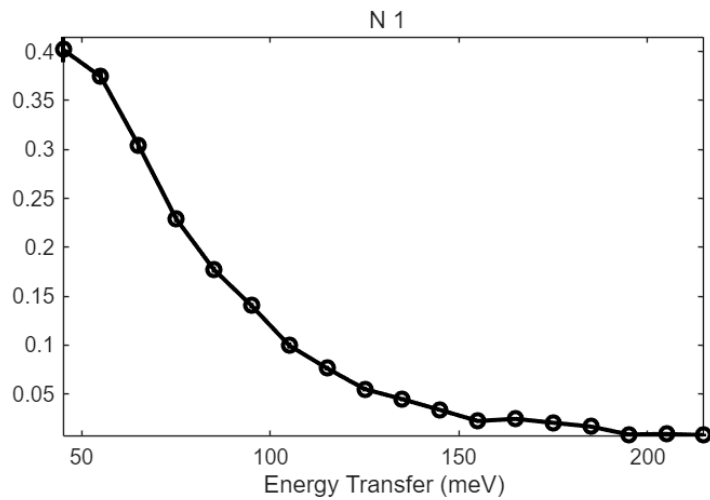
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

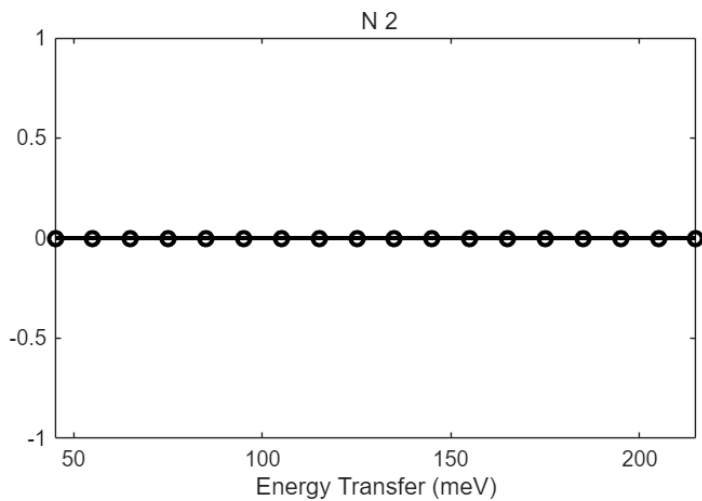
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
```

%end